

Software Developers (O*NET-SOC Code 15-1252.00)

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Overview

Research, design, and develop computer and network software or specialized utility programs. Analyze user needs and develop software solutions, applying principles and techniques of computer science, engineering, and mathematical analysis. Update software or enhance existing software capabilities. May work with computer hardware engineers to integrate hardware and software systems, and develop specifications and performance requirements.

Tasks

1. Analyze user needs and software requirements to determine feasibility of design within time and cost constraints.
2. Develop or direct software system testing or validation procedures, programming, or documentation.
3. Confer with systems analysts, engineers, programmers and others to design systems and to obtain information on project limitations and capabilities.
4. Modify existing software to correct errors, adapt it to new hardware, or upgrade interfaces and improve performance.
5. Prepare reports or correspondence concerning project specifications, activities, or status.

Essential Skills

- Critical Thinking - Using logic and reasoning to identify the strengths and weaknesses of alternative solutions.
- Active Learning - Understanding the implications of new information for current and future problem-solving.
- Reading Comprehension - Understanding written sentences and paragraphs in work-related documents.
- Active Listening - Giving full attention to what other people are saying.
- Writing - Communicating effectively in writing as appropriate for the needs of the audience.
- Speaking - Talking to others to convey information effectively.
- Monitoring - Assessing performance of yourself, other individuals, or organizations to make improvements.

Knowledge Areas

- Computers and Electronics - Knowledge of circuit boards, processors, and computer hardware and software.

- Customer and Personal Service - Knowledge of principles and processes for providing customer services.
- Mathematics - Knowledge of arithmetic, algebra, geometry, calculus, and statistics.
- English Language - Knowledge of the structure and content of the English language.

Work Styles

- Innovation - A tendency to be inventive and adopt new perspectives on ways to accomplish work.
- Adaptability - A tendency to be open to and comfortable with change, new experiences, or ideas at work.
- Achievement Orientation - A tendency to establish and maintain personally challenging work-related goals.
- Intellectual Curiosity - A tendency to seek out new work-related knowledge.
- Attention to Detail - A tendency to be detail-oriented, organized, and thorough in completing work.
- Dependability - A tendency to be reliable, responsible, and consistent in meeting work-related obligations.

Education

Most respondents reported that a Bachelor's degree is required to perform a job in this occupation. A smaller share reported that an Associate's degree or a Master's degree is required.

Wages and Employment Trends

Median wages (2025): \$65.38 hourly, \$135,980 annual. Employment (2024): 1,693,800 employees. Projected growth (2024-2034): much faster than average.